

TOPIC 7: FACTORIZATION

1. Common Factor

$ax + ay = a(x + y)$. Always take out the highest common factor first.

2. Grouping

$3m + pq + 3p + mp = (3m + mp) + (3p + pq) = m(3 + p) + p(3 + q)$ — regroup to get a common binomial: e.g. $3m + 3p + mp + pq = 3(m + p) + p(m + q) = (m + p)(3 + q)$ when the pairs match. Practise rearranging terms so that grouping works.

3. Difference of Two Squares

$a^2 - b^2 = (a - b)(a + b)$. Example: $x^2 - 9 = (x - 3)(x + 3)$.

4. Perfect Square Trinomial (related form)

$a^2 - 2ab + b^2 = (a - b)^2$; $a^2 + 2ab + b^2 = (a + b)^2$. Recognising these speeds up factorisation.

5. Word Problems

Translate the problem into an algebraic expression, factorise if it simplifies the equation, then solve. Check answers in the original context.

6. Summary

Factorisation rewrites an expression as a product. Use common factors, grouping, difference of squares and perfect squares as tools.

End of Topic 7